Historical Stock Analysis

Does diversification survive a closer look?

230,111
ticker-days

49
global tickers

20 years
2006–2026

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What we analyzed and how

230,111 daily company records; Python tested the hypotheses and R replicated the monthly analysis.

230,111 rows | 8 columns | 0 missing values | 2006-01-02 → 2026-02-20

Date	Ticker	Open	High	Low	Close	Adj Close	Volume
2026-02-20	AAPL	258.97	264.75	258.16	264.58	264.58	4,20,44,900
2026-02-20	005930.KS	1,90,000	1,90,300	1,88,600	1,90,100	1,90,100	2,42,13,880
2026-02-20	MC.PA	541.40	559.20	540.10	554.70	554.70	9,29,807

Open, High, Low, and Close describe the daily price range; Adj Close includes dividend adjustments; Volume is shares traded.

Three hypotheses guide the analysis

Each hypothesis tests one part of the same diversification question.

01 H1 · AI-rally concentration

The selected chip/AI group will outperform the selected broader-tech group during the AI rally.

02 H2 · Regional diversification during crises

Average regional correlations will be higher in crisis periods than in calm years.

03 H3 · Volatility and volume during panic periods

Volatility and trading volume will rise during panic periods relative to the calm baseline.

Together, H1 tests concentration; H2 and H3 test whether diversification weakens under stress.

How we test the three hypotheses

One technology comparison and three crisis measures, using adjusted prices.

TECHNOLOGY GROUPS

Chip / AI

NVDA · AVGO · TSM · ASML · AMD · MU · LRCX

Broader tech

AAPL · MSFT · GOOGL · AMZN · META · NFLX · ORCL · CSCO

Indexed value = 100 × Adj Close / starting Adj Close

MARKET PERIODS

Calm baseline

2015–2017 · no major global shock

2008 crisis

Housing collapse, mortgage losses, and a global banking shock

2020 crash

Pandemic lockdowns triggered a sudden global selloff

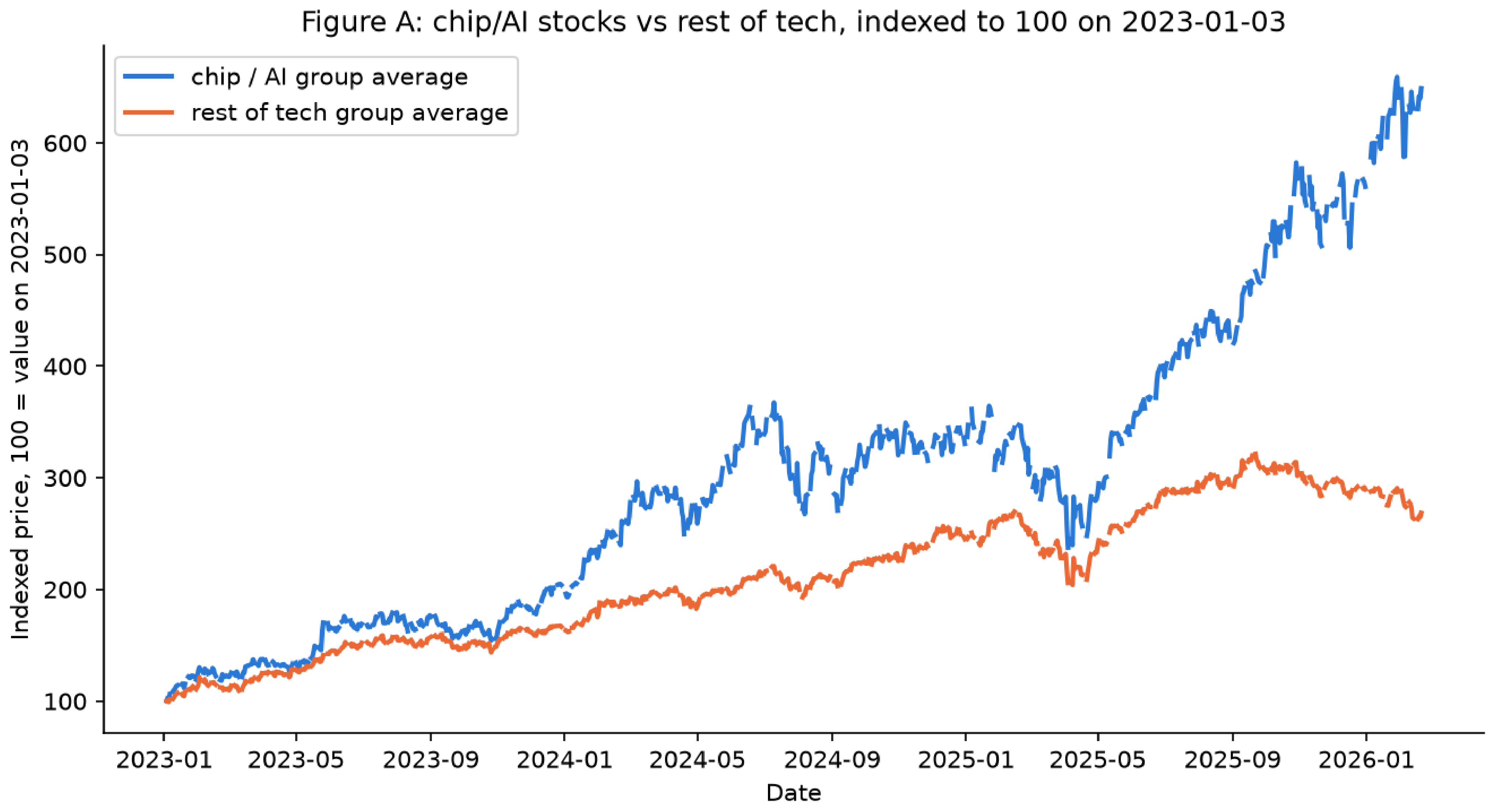
2022 drawdown

Inflation and rapid rate hikes drove a slower drawdown

Measures: correlation, volatility, and volume relative to calm

The selected chip/AI group pulled decisively ahead

Indexed group averages from 2023-01-03 to 2026-02-20.



Source: data/stock_data.csv, 2023-01-03 to 2026-02-20. Chip/AI group: NVDA, AVGO, TSM, ASML, AMD, MU, LRCX.
Rest of tech group: AAPL, MSFT, GOOGL, AMZN, META, NFLX, ORCL, CSCO. Each group line is a simple average of the indexed prices.

+549%

chip / AI group

+168%

broader-tech group

0.26% vs 0.14%

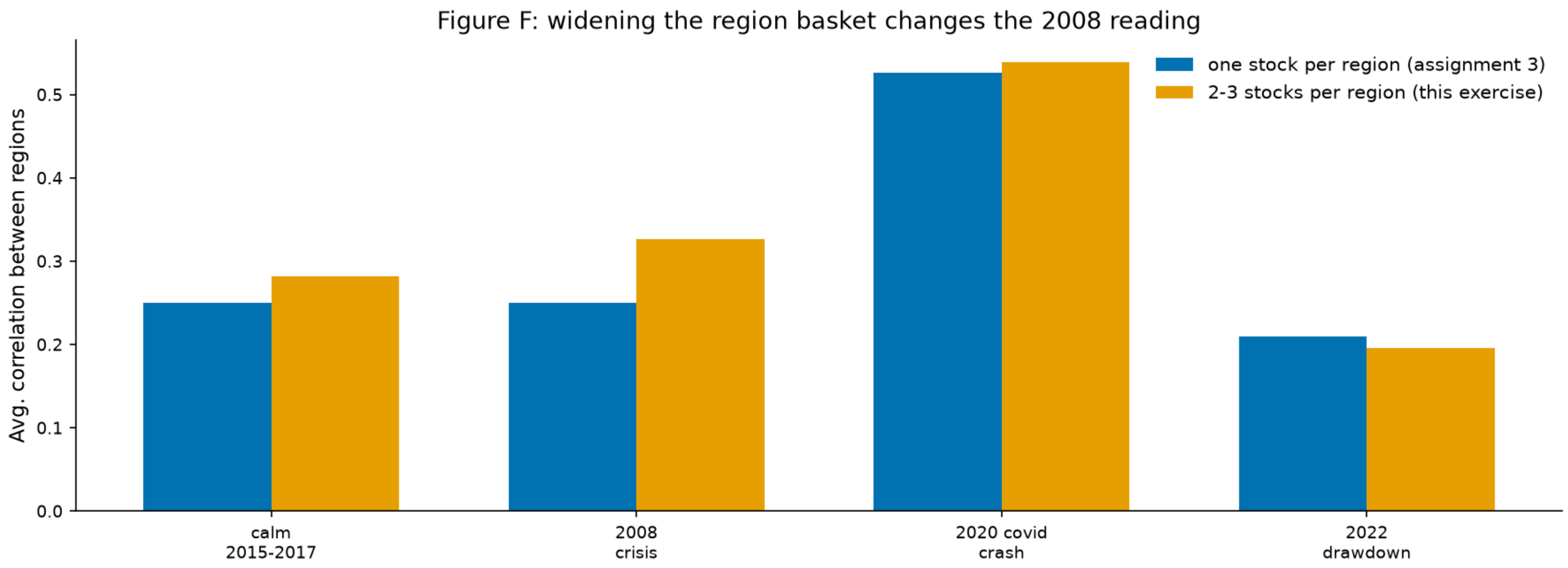
mean daily return

Welch p = 0.0073

Conclusion: H1 is supported for these selected groups—not for every company linked to AI.

Regional diversification weakened most in the 2020 crash

Average pairwise correlations after widening the available regional baskets.



0.282
calm

0.326
2008

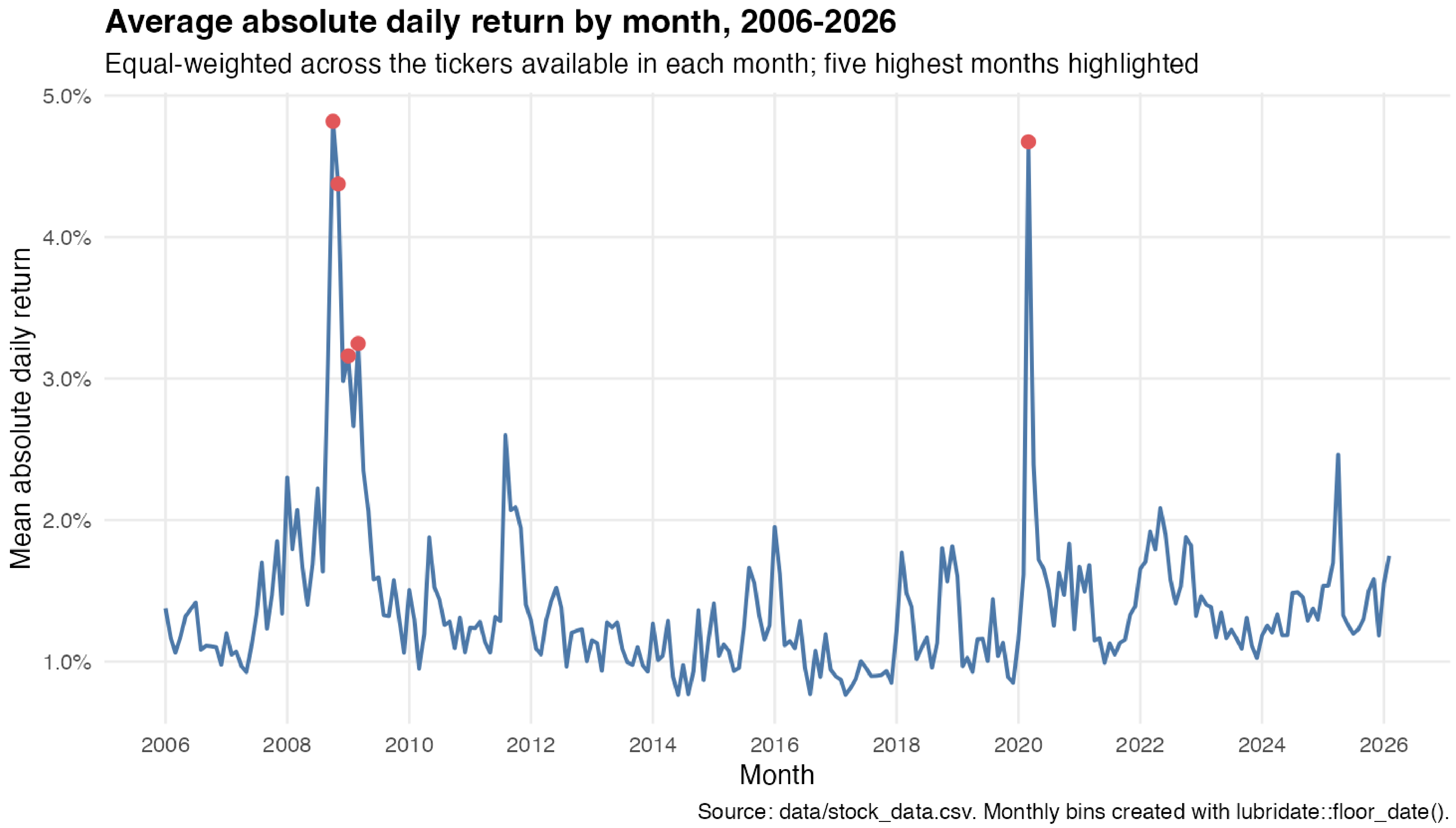
0.539
2020

The 2008 estimate became more crisis-like after widening the basket, but Paris still has one stock and Korea/Switzerland remain sector-narrow.

Conclusion: H2 is strongly supported for 2020 and directionally supported for 2008.

Panic months produced the largest volatility spikes

Equal-weighted mean absolute daily return by month across available tickers.



OCT 2008

4.82%

MAR 2020

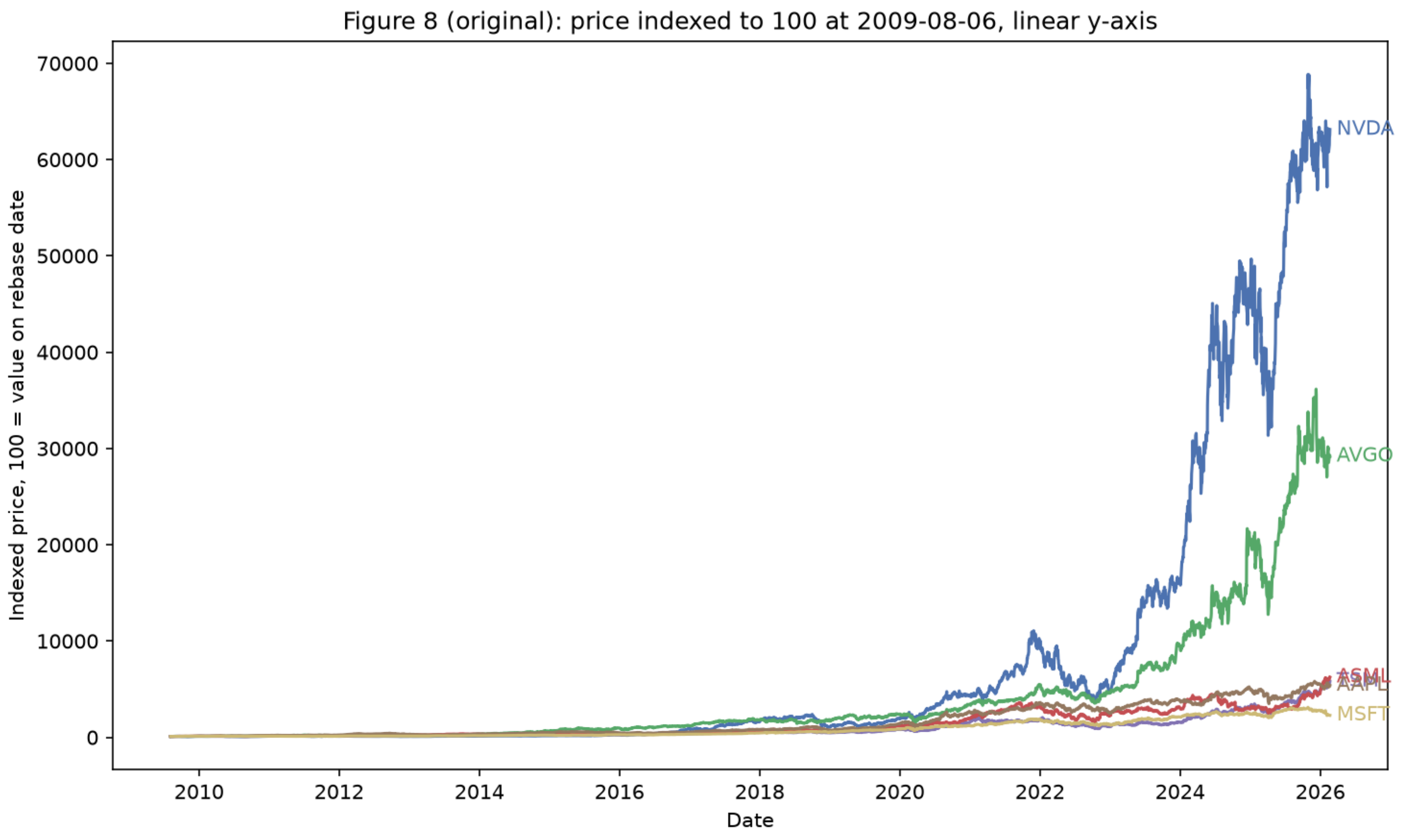
4.67%

Independent monthly aggregation reaches the same crisis conclusion.

A log scale reveals the growth stories hidden by the linear chart

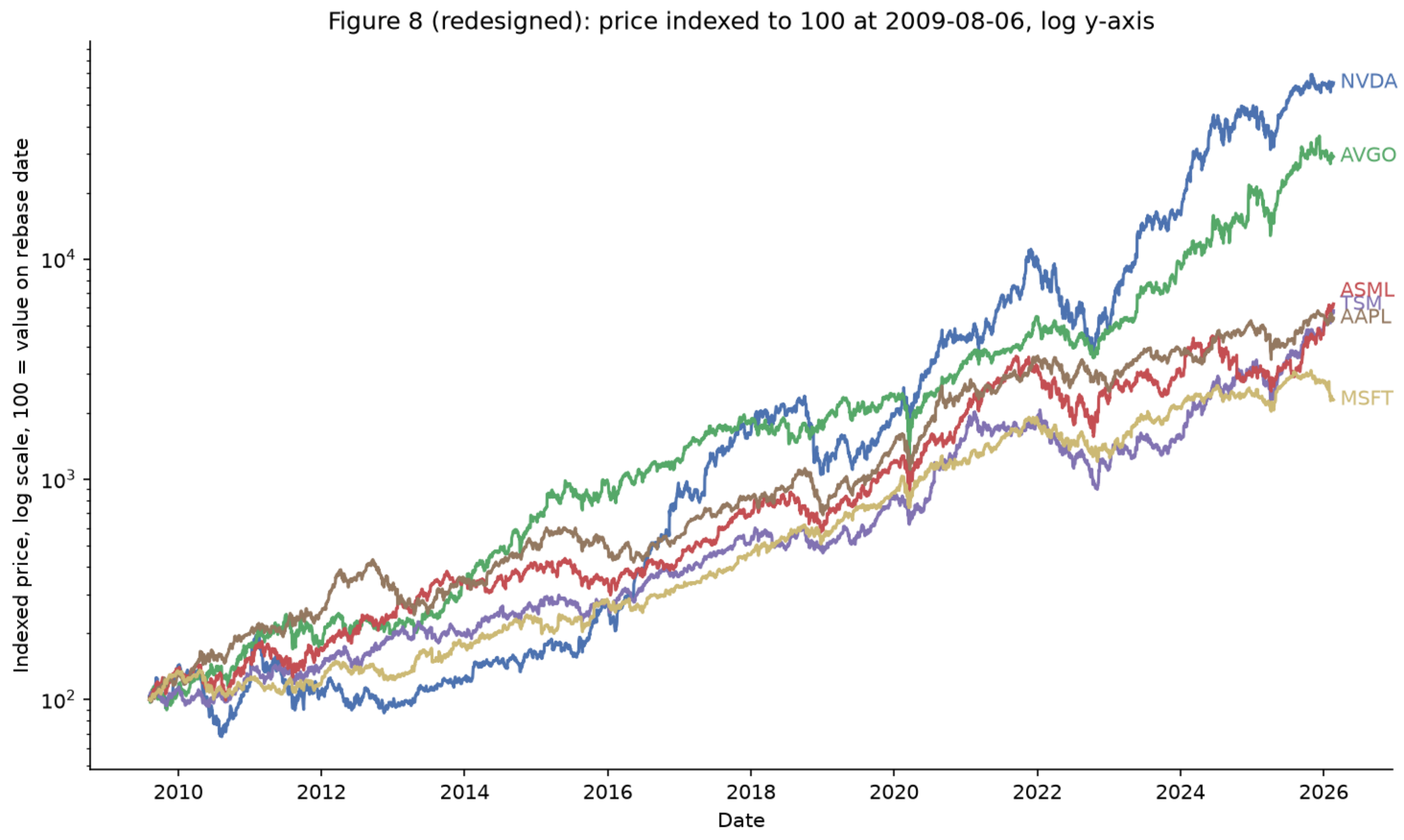
Same data and rebasing; only the visual encoding changes.

BEFORE · LINEAR



NVDA and AVGO dominate the absolute range, compressing four lines near the baseline.

AFTER · LOG



Equal vertical distance now represents equal proportional change; all six series remain legible.

Principles applied: appropriate scale · direct labels · less non-data ink · better pre-attentive separation

Diversification is thinner when it matters most

All three hypotheses support the central story, within clear limits.

01 Technology leadership was concentrated

The selected chip/AI group gained 549% versus 168% for broader tech.

02 Regional protection weakened in sudden stress

Average correlation peaked at 0.539 in the 2020 crash.

03 Volatility independently confirmed the panic signal

October 2008 and March 2020 were the largest monthly spikes.

What we can claim: patterns in this dataset. What we cannot claim: causality, forecasting, or a universal market law.

Make the evidence explorable

The dashboard keeps every assumption visible as a filter.

POWER BI / TABLEAU

One overview

Filter by ticker group, region, and crisis period; compare indexed performance, correlations, and monthly volatility.

R WRANGLING

Long-form by design

Use parsed dates, month bins, and one Ticker field so color, grouping, and filters remain consistent.

OPEN RISKS

unequal regional coverage · share-count volume · removed original Kaggle page · no causal interpretation

Success means the user can reproduce each headline and see where the evidence becomes uncertain.